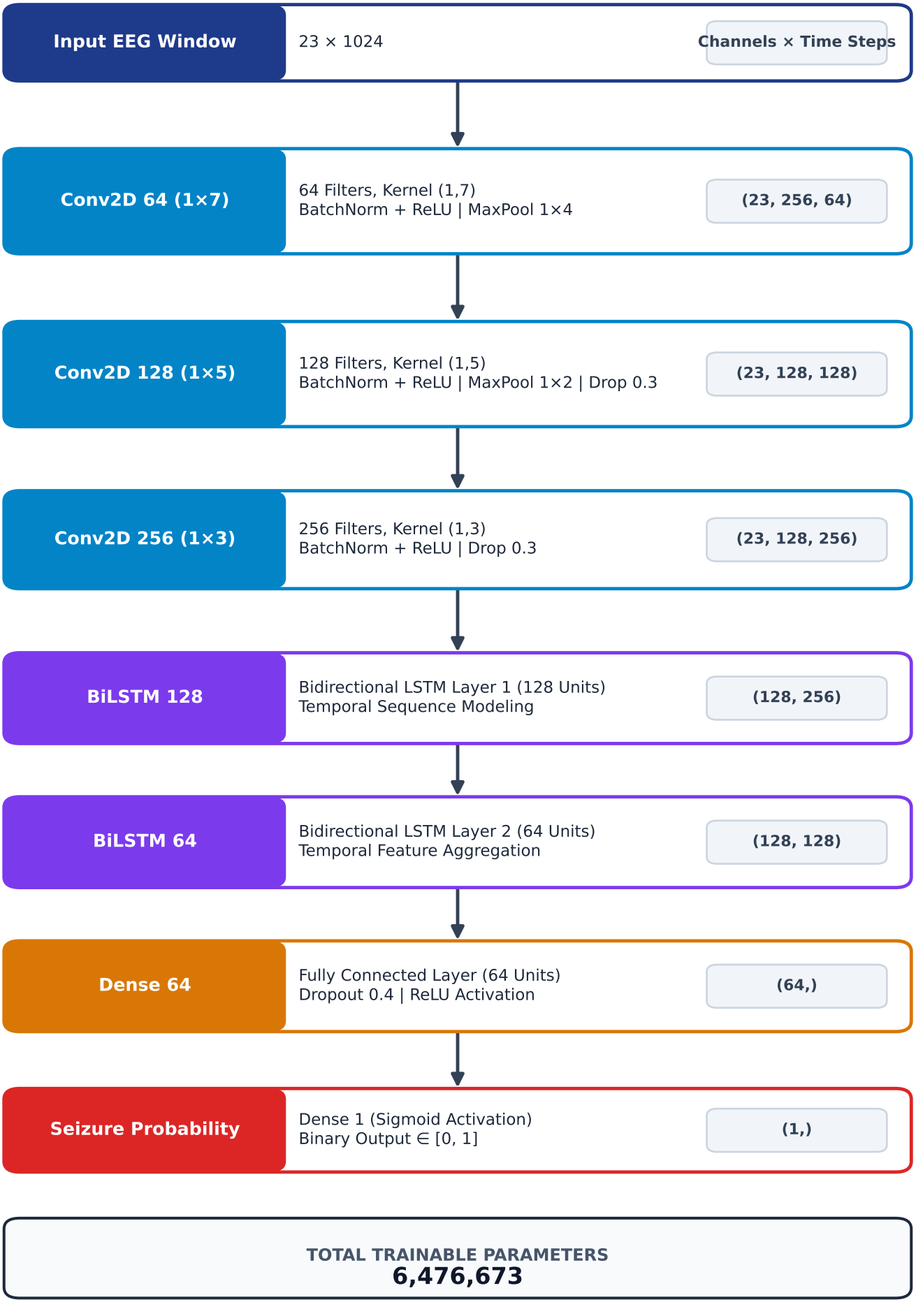
CNN–BiLSTM NEURAL NETWORK ARCHITECTURE
Input EEG Window
23 × 1024
Channels × Time Steps
Conv2D 64 (1×7)
64 Filters, Kernel (1,7)
BatchNorm + ReLU | MaxPool 1×4
(23, 256, 64)
Conv2D 128 (1×5)
128 Filters, Kernel (1,5)
BatchNorm + ReLU | MaxPool 1×2 | Drop 0.3
(23, 128, 128)
Conv2D 256 (1×3)
256 Filters, Kernel (1,3)
BatchNorm + ReLU | Drop 0.3
(23, 128, 256)
BiLSTM 128
Bidirectional LSTM Layer 1 (128 Units)
Temporal Sequence Modeling
(128, 256)
BiLSTM 64
Bidirectional LSTM Layer 2 (64 Units)
Temporal Feature Aggregation
(128, 128)
Dense 64
Fully Connected Layer (64 Units)
Dropout 0.4 | ReLU Activation
(64,)
Seizure Probability
Dense 1 (Sigmoid Activation)
Binary Output ∈ [0, 1]
(1,)
TOTAL TRAINABLE PARAMETERS
6,476,673